	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 8

Lab Title: **Priority Queues**

Student Name: **Muammad Waqas** **Reg. No:** **240756**

Objective:. Understand the concept of riority queue and how it follows the FIFO (First In, First Out) **.** principle. Implement a queue using a linked list without using arrays or STL containers

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

Department of Computer Engineering

Data Structures And Algorithms

LAB REPORT : 8

Tittle : Priority queue

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Submitted To : CE . Engr Maryam

Date : 23 th November 24

Objective

The main objective of this program is to:

- Understand the concept of a queue and how it follows the FIFO (First In, First Out) principle.
- Implement a queue using a linked list without using arrays or STL containers.
- Learn how to use pointers to dynamically create and connect nodes.
- Practice basic enqueue (insert), dequeue (delete), and display operations using classes and objects in C++.

TASK 1

1. At an airport check-in counter, passengers line up to board in the order they arrive.
However, Business Class passengers are always given priority over Economy Class passengers..

====(CODE)====

```
#include<iostream>
using namespace std;

struct node{
string n,country,cls;
int id;
node* next;
};

class queueLL{
public:
node *f,*r,*lb;
queueLL(){f=r=lb=NULL;}

bool empty(){return f==NULL;}

void enq(string nm,int i,string ct,string c)
node* t=new node();
t->n=nm;
t->id=i;
t->country=ct;
t->cls=c;
t->next=NULL;

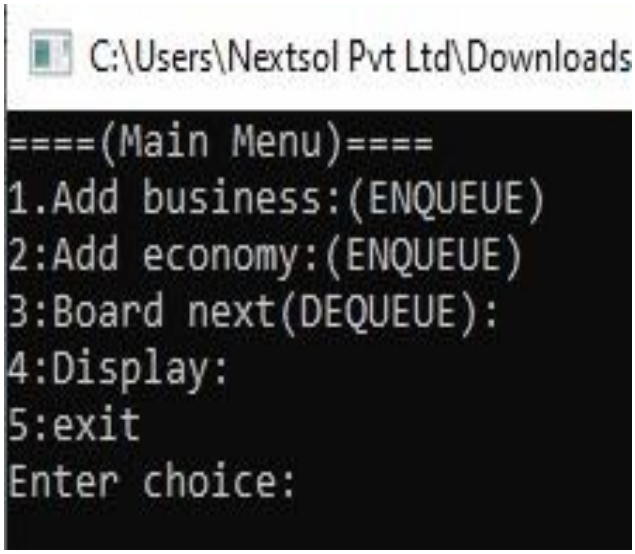
if(empty()){
f=r=t;
if(c=="B")lb=t;
return;
}
if(c=="B"){
if(lb==NULL){
t->next=f;
f=t;
lb=t;}
else{
t->next=lb->next;
lb->next=t;
lb=t;
if(lb->next==NULL)r=lb;
}}
else{
r->next=t;
r=t;
}}
void deq(){
if(empty()){cout<<"queue empty\n";return;
node* t=f;
cout<<"boarding:"<<t->n<<
" id:"<<t->id<<" "<<
t->country<<"("<<t->cls<<")\n";
f=f->next;
if(t==lb)lb=NULL;
if(f==NULL)r=NULL;
delete t;
}
```

====(CODE)====

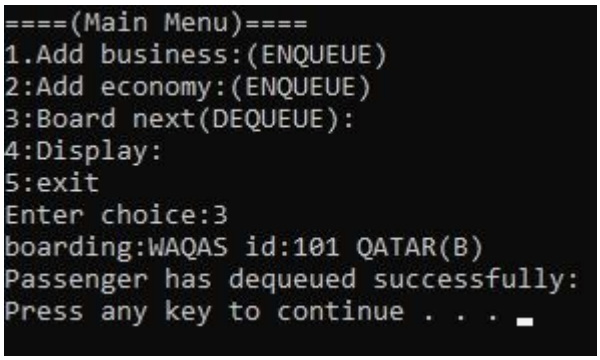
```
56  void show(){
57  if(empty()){cout<<"empty\n";return;}
58  node* t=f;
59  int n=0;
60  while(t!=NULL){
61  cout<<"===(PASSENGER "<<(++n)
62  <<" CLASS: "<<t->cls<<")====="<<endl;
63  cout<<"Name: "<<t->n<<endl;
64  cout<<"ID_CARD:"<<t->id<<endl;
65  cout<<"Country "<<t->country<<endl;
66  cout<<"CLASS: "<<t->cls<<endl;
67  t=t->next;
68  }
69  cout<<"NULL\n";
70  }};
71  int main(){
72  queueLL q;
73  int ch,id;
74  string nm,ct,c;
75  do{
76  system("cls");
77  cout<<"====(Main Menu)====\n";
78  cout<<"1.Add business:(ENQUEUE)"<<endl
79  cout<<"2:Add economy:(ENQUEUE)"<<endl;
80  cout<<"3:Board next(DEQUEUE):"<<endl;
81  cout<<"4:Display:"<<endl;
82  cout<<"5:exit\n";
83  cout<<"Enter choice:";
84  cin>>ch;
85  if(ch==1){
86  cout<<"name:";cin>>nm;
87  cout<<"id:";cin>>id;
88  cout<<"country:";cin>>ct;
89  q.enq(nm,id,ct,"B");
90  }
91  else if(ch==2){
92  cout<<"name:";cin>>nm;
93  cout<<"id:";cin>>id;
94  cout<<"country:";cin>>ct;
95  q.enq(nm,id,ct,"E");
96  }
97  else if(ch==3){
98  q.deq();
99  cout<<"Passenger has dequeued successfully:\n";
100 system("pause");
101 }
102 else if(ch==4){
103 q.show();
104 system("pause");
105 }
106 else if(ch!=5){
107 cout<<"invalid\n";
108 system("pause");
109 }
110 }while(ch!=5);
111 return 0;
112 }
```

====(OUTPUT)====

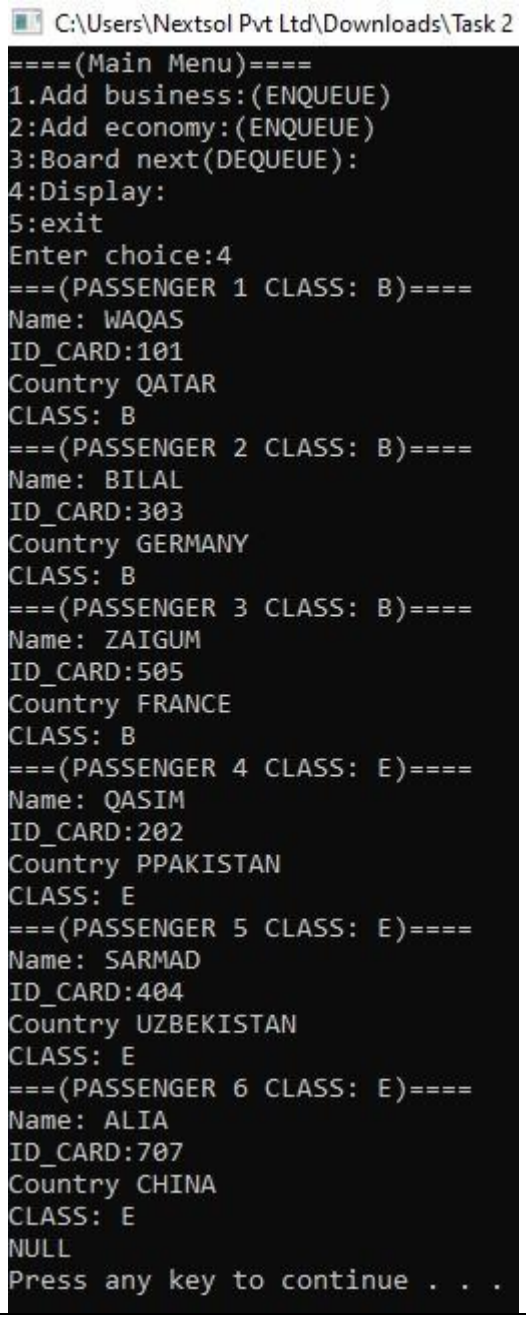
MENU
ENET BUSINESS CLAS BY 1
ENETR ECONOMY CLASS BY 2



SERVED TE PASSENGER



DISPLAY PASSENGERS QUEUE



TASK 2\

2. A software company handles customer support requests. Requests are categorized:

- ✓ **Premium customers (high priority)**
- ✓ **Standard customers (low priority)**
- ✓ **Requests must be handled as follows:**
- ✓ **Premium customer requests always get served first**

====(CODE)====

```
#include<iostream>
using namespace std;
class node{
    public:
string n;
int token;
node* next;
};
class queueLL{
public:
node *f,*r;
queueLL(){f=r=NULL;}
bool empty(){return f==NULL;}
void enq(string nm,int tk){
node* t=new node();
t->n=nm;
t->token=tk;
t->next=NULL;
if(empty()){
f=r=t;
}
else{
r->next=t;
r=t;}
}
void deq(){
if(empty()){cout<<"Queue empty\n";return;
node* t=f;
cout<<"Serving:"<<t->n
<<" Token:"<<t->token<<"\n";
f=f->next;
if(f==NULL)r=NULL;
delete t;
}
void show(){
if(empty()){cout<<"Empty\n";return;}
node* t=f;
int c=0;
while(t!=NULL){
cout<<"===(REQUEST "<<(++c)<<")====\n"
cout<<"Name:"<<t->n<<endl;
cout<<"Token:"<<t->token<<endl;
t=t->next;
}}};
int main(){
queueLL premium,standard;
int ch,tk;
string nm;
do{
system("cls");
cout<<"====(Main Menu)====\n";
cout<<"1.Add Premium (ENQUEUE)\n";
cout<<"2.Add Standard (ENQUEUE)\n";
cout<<"3.Serve Next (DEQUEUE)\n";
cout<<"4.Display\n";
cout<<"5.Exit\n";
```

====(CODE)====

```
57  cout<<"Enter choice:";
58  cin>>ch;
59  if(ch==1){
60  cout<<"Name:";cin>>nm;
61  cout<<"Token:";cin>>tk;
62  premium.enq(nm,tk);
63  }
64  else if(ch==2){
65  cout<<"Name:";cin>>nm;
66  cout<<"Token:";cin>>tk;
67  standard.enq(nm,tk);
68  }
69  else if(ch==3){
70  cout<<"----Serving Request----\n";
71  if(!premium.empty()){
72  premium.deq();
73  }
74  else if(!standard.empty()){
75  standard.deq();
76  }
77  else{
78  cout<<"No pending requests.\n";
79  }
80  system("pause");
81  }
82  else if(ch==4){
83  cout<<"\n----Premium Queue----\n";
84  premium.show();
85  cout<<"\n----Standard Queue----\n";
86  standard.show();
87  cout<<endl;
88  system("pause");
89  }
90  else if(ch!=5){
91  cout<<"Invalid\n";
92  system("pause");}
93  }while(ch!=5);
94  return 0;
95  }
```

====(OUTPUT)====

MAIN MENU

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```
====(Main Menu)====
1.Add Premium (ENQUEUE)
2.Add Standard (ENQUEUE)
3.Serve Next (DEQUEUE)
4.Display
5.Exit
Enter choice:4
```

SERVING REQUEST FISRT PREMIUM AND THEN STADARD REQUEST:

C:\Users\Nextsol Pvt Ltd\Downloads\Untitle

```
====(Main Menu)====
1.Add Premium (ENQUEUE)
2.Add Standard (ENQUEUE)
3.Serve Next (DEQUEUE)
4.Display
5.Exit
Enter choice:3
----Serving Request----
Serving:A Token:1
Press any key to continue . . .
```

AFTER END OF PREMIUM REQUEST THEN STADAR QUEUE TURN COME:

C:\Users\Nextsol Pvt Ltd\Downloads\Untitled1.exe

```
====(Main Menu)====
1.Add Premium (ENQUEUE)
2.Add Standard (ENQUEUE)
3.Serve Next (DEQUEUE)
4.Display
5.Exit
Enter choice:4

----Premium Queue----
Empty

----Standard Queue----
===(REQUEST 1)====
Name:B
Token:2
===(REQUEST 2)====
Name:D
Token:4
===(REQUEST 3)====
Name:R
Token:6

Press any key to continue . . .
```

ADD DATA AND DISPLAY:

C:\Users\Nextsol Pvt Ltd\Downloads\Untitled1

```
====(Main Menu)====
1.Add Premium (ENQUEUE)
2.Add Standard (ENQUEUE)
3.Serve Next (DEQUEUE)
4.Display
5.Exit
Enter choice:4

----Premium Queue----
===(REQUEST 1)====
Name:A
Token:1
===(REQUEST 2)====
Name:C
Token:3
===(REQUEST 3)====
Name:5
Token:5

----Standard Queue----
===(REQUEST 1)====
Name:B
Token:2
===(REQUEST 2)====
Name:D
Token:4
===(REQUEST 3)====
Name:R
Token:6

Press any key to continue . . .
```

Conclusion:

A priority queue is a special kind of queue where items are not served strictly in the order they arrive, but according to their priority. This means important tasks or people like Business Class passengers or emergency cases are handled first. Understanding priority queues helps us solve real-life problems where urgency matters more than time of arrival. They are widely used in hospital emergency rooms, airport check-ins, operating systems for process scheduling, and network packet handling. Learning how priority queues work improves logical thinking and teaches us how data can be managed efficiently based on importance rather than just arrival order.